Effect of Butter Lamination and Resting Time on Crumb Structure and Sensory Quality of Croissants

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Abstract

Laminated pastries such as croissants owe their open, flaky crumb to alternating layers of dough and butter. This study evaluated how the number of lamination folds and the resting time between folds affect crumb structure and sensory quality. We produced croissants using three fold schemes (two, three and four letter folds) and two resting times (20 and 45 minutes) in a controlled bakery. Crumb structure was quantified by image analysis of cross-sections, and a trained panel of twelve assessors scored flakiness, tenderness and butter flavor. Three folds combined with a 45-minute rest produced the most open crumb and the highest panel scores. Four folds caused butter to merge with the dough and reduced layer definition. We conclude that moderate lamination with adequate rest gives the best balance of structure and eating quality.

1. Introduction

The quality of a croissant depends on the number, thickness and continuity of its butter layers. During baking, water in the butter turns to steam and lifts the dough layers apart, producing the characteristic honeycomb crumb. If the butter melts into the dough or the layers tear, steam escapes and the pastry bakes dense.

Bakers typically choose the number of folds and the rest time by experience. This study aimed to quantify their effect on crumb structure and on the perception of quality by trained tasters.

2. Methods

Dough and butter. All batches used the same recipe: strong bread flour, 12% sugar, 2% salt, 1.5% instant yeast and 25% of dough weight as laminating butter with 82% fat. Butter was kept at 14 degrees C before lamination.

Treatments. Three fold schemes (two, three or four letter folds) were crossed with two rest times between folds (20 or 45 minutes) in a refrigerated room at 4 degrees C, giving six treatments with five croissants each.

Measurements. Baked croissants were cut in half and photographed. Image analysis counted the number of alveoli and measured the mean cell area. A panel of twelve trained assessors scored flakiness, tenderness and butter flavor on a nine-point scale.

3. Results

Three folds with a 45-minute rest produced the highest number of distinct alveoli (mean 118 per cross-section) and the largest mean cell area. Four folds reduced alveoli to 86 per cross-section, and photographs showed fused layers where butter had been absorbed by the dough. Two folds gave thick, uneven layers.

Short rests of 20 minutes increased butter leakage during baking in all fold schemes. Panel scores for flakiness were highest for three folds with a 45-minute rest (mean 7.6) and lowest for four folds with a

20-minute rest (mean 5.2). Butter flavor was rated similarly across treatments.

4. Discussion

The findings suggest that more folds do not always mean better layers. Beyond a certain point the layers become too thin to remain separate, and the mechanical work warms the butter. A longer rest lets the gluten relax and keeps the butter firm, which protects the layers during rolling.

These observations agree with common bakery practice, but the measurements give bakers numbers they can use to standardize production.

5. Limitations

Only one flour, one butter and one oven were used, so the results may not transfer to other ingredients or equipment. The panel was small and trained on this product, which may not represent consumers. Room temperature and dough temperature were controlled but not varied. We did not test frozen storage or proofing time.

6. Conclusion

Three letter folds with a 45-minute rest gave the most open crumb and the highest sensory scores. Future work should test other fat types and proofing conditions.

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